

Title: STRUCTURE AND FUNCTION OF THE HUMAN BODY

National Cadet Corps (NCC) â Official Directorate General Training Notes

1. **LEARNING OBJECTIVES & OVERVIEW:** The human body is a highly coordinated biological system consisting of nine major organ systems working harmoniously: Skeletal, Muscular, Circulatory, Respiratory, Digestive, Excretory, Nervous, Endocrine, and Reproductive systems. Physical military readiness requires a comprehensive understanding of physiological limits and anatomical conditioning.

2. **SKELETAL & MUSCULAR SYSTEMS:** The human skeletal framework is composed of 206 bones. It serves three critical functions: providing structural rigidity, protecting delicate internal organs (such as the brain inside the cranium and heart/lungs within the thoracic cage), and acting as mechanical levers for voluntary muscle groups. Physical training (PT), cross-country marching, and obstacle courses build high bone mineral density and muscular endurance.

3. **CIRCULATORY & RESPIRATORY SYSTEMS:** The cardiovascular engine is powered by the four-chambered heart, circulating approximately 5 liters of oxygenated blood per minute. The respiratory system exchanges oxygen and carbon dioxide through 300 million pulmonary alveoli. Endurance marching enhances maximum oxygen uptake (VO_2 max), allowing cadets to march 20 km under full battle load without acute lactic acidosis.

4. **INSTRUCTOR FIELD SUMMARY:** Regular physical conditioning, proper hydration, balanced nutritional intake, and adequate sleep recovery are essential to maintain unit operational readiness.